



Image by co-reasoning: A collaborative reasoning-based implicit data augmentation method for dual-view CEUS classification

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ABSTRACT

Dual-view contrast-enhanced ultrasound (CEUS) data are often insufficient to train reliable machine learning models in typical clinical scenarios. A key issue is that limited clinical CEUS data fail to cover the underlying texture variations for specific diseases. Implicit data augmentation offers a flexible way to enrich sample diversity, however, inter-view semantic consistency has not been considered in previous studies. To address this issue, we propose a novel implicit data augmentation method for dual-view CEUS classification, which performs a sample-adaptive data augmentation with collaborative semantic reasoning across views. Specifically, the method constructs a feature augmentation distribution for each ultrasound view of an individual sample, accounting for intra-class variance. To maintain semantic consistency between the augmented views, plausible semantic changes in one view are transferred from similar instances in the other view. In this retrospective study, we validate the proposed method on the dual-view CEUS datasets of breast cancer and liver cancer, obtaining the superior mean diagnostic accuracy of 89.25% and 95.57%, respectively. Experimental results demonstrate its effectiveness in improving model performance with limited clinical CEUS data. Code: <https://github.com/wanpeng16/CRIDA>.

1. Introduction

Dual-view contrast-enhanced ultrasound (CEUS) that combines B-mode ultrasound (BUS), offers a comprehensive imaging approach for a variety of clinical tasks, such as disease screening, malignancy grading, and prognosis evaluation. As depicted in Fig. 1, BUS captures lesion morphological characteristics like shape, size, margin, and echo patterns. Dynamic CEUS allows for a real-time assessment of tumor blood supply, where diverse enhancement patterns reflect the degree of enhancement, homogeneity, and blood flow velocity, which are indicative of tumor functional characteristics, such as micro-vessel distribution, tissue necrosis, and the extent of tumor invasion (Bartolotta et al., 2019; Radzina et al., 2021; Minami and Kudo, 2011). Recently, various machine learning algorithms are proposed to automate the imaging interpretation process, pursuing an objective and efficient aided diagnosis.

Previous studies generally formulate dual-view CEUS analysis as a multi-view learning (Zhang et al., 2023a; Han et al., 2023) or video

classification problem (Liang et al., 2016; Chen et al., 2021), emphasizing the aspect of multi-view fusion or dynamic feature extraction. While these methods have shown promising results in dual-view CEUS classification, their effectiveness largely hinges on the availability of large-scale annotated datasets, which guarantees to capture stable and robust correlations between images and labels. However, in practical deployment, collecting sufficient dual-view CEUS data is particularly challenging due to the time-consuming and costly data acquisition, as well as the need of expert annotations. A common problem is that collected dual-view CEUS data are confounded by data acquisition condition, considering the variability in imaging protocols and patient conditions.

Data augmentation is an effective approach to enrich training samples and reduce the risk of over-fitting (Wang et al., 2019; Hong et al., 2022; Volpi et al., 2018; Li et al., 2021). A typical solution is to generate artificial samples with controlled semantic changes using

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deep generative models (Chadebec et al., 2022; Frid-Adar et al., 2018; Dalmaz et al., 2023). However, these generative models inherently demand substantial training data, which is inconsistent with the real-world clinical settings where data are scarce. An alternative approach is implicit data augmentation (Wang et al., 2019), which perturbs training samples in the latent representation space. Owing to the property that deep representation spaces are well linearized (Upchurch et al., 2017), shifting samples along specific directions induces corresponding semantic changes.

However, existing implicit data augmentation methods are typically designed for single-modality scenarios, where semantic perturbations are reasoned independently for each modality without considering inter-modality semantic consistency. We argue that radiologists can image plausible variations of the original case based on their prior knowledge, for example, *how tissue echos might vary given tumor perfusion pattern, or which type of enhancement texture might appear given tumor morphology*. That is, augmented US views are supposed to be aligned in the feature space, and consistently reflect the morphological and functional characteristics of the same lesion. Another limitation is that previous studies generally sample semantic directions from class-wise augmentation distributions, ignoring sample variances in the same class (Chen et al., 2022). Considering the fact that dual-view CEUS in the same class often exhibit large imaging variability, we argue that the optimal augmentation distribution should be *sample-adaptive* to perform semantic perturbation flexibly.

Inspired by this, we propose a Collaborative Reasoning-based Implicit Data Augmentation (CRIDA) method for dual-view CEUS classification. CRIDA generates plausible augmented data through cross-view collaborative reasoning, reducing the risk of over-fitting under limited dual-view CEUS data. This method augments training samples by translating them along certain semantic directions, which are sampled from a sample-adaptive augmentation distribution, with its covariance matrix estimated based on feature statistics of training data. To ensure that augmented BUS and CEUS view are semantically aligned, our method follows a two-step process. (1) We first train the network with the available dual-view CEUS data and cluster training samples into K groups independently for each view together with category labels. Then, covariance matrix for each cluster is estimated to capture potential semantic variations within that cluster. (2) We then construct sample-adaptive augmentation distribution by the manner of collaborative reasoning (as shown in Fig. 1). For example, when addressing the question *which type of enhancement texture might appear given tumor morphology*, we search the transferable enhancement pattern changes from similar clusters in the BUS view, borrowing semantic variations from instances with similar tumor morphology. Similarly, when considering *how tissue echos might vary given tumor perfusion pattern*, we explore the transferable echos changes from similar clusters in the CEUS view, propagating transformation directions from instances with comparable enhancement patterns. The main contributions of our CRIDA method are summarized as three folds.

- We propose a sample-adaptive implicit data augmentation method, which constructs feature augmentation distribution for each US view of individual sample. This approach considers the sample variance of the same disease class, offering the greater flexibility for semantic perturbation.
- We introduce a collaborative reasoning strategy to maintain semantic alignment between augmented BUS and CEUS views. This strategy explores transferable semantic directions of one view from similar instances in the other view, thus consistently reflecting morphological and functional characteristics of the same lesion.
- We evaluate our CRIDA method on the two dual-view CEUS datasets of breast cancer and liver cancer. Our method achieves the superior diagnostic accuracy of 89.25% and 95.57%, respectively, demonstrating its efficacy in improving model performance with boosted sample diversity.

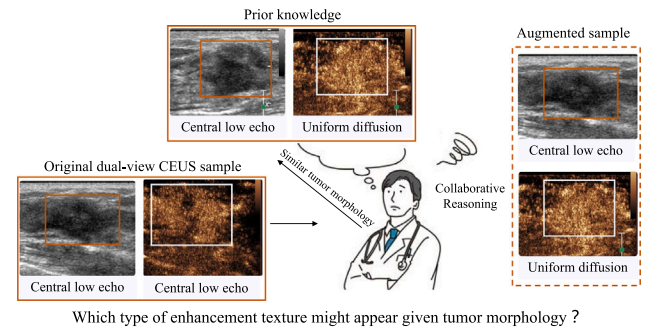


Fig. 1. An illustration of collaborative semantic reasoning for implicit data augmentation. We could reason the possible enhancement pattern (i.e., uniform diffusion) of CEUS view by the same echo pattern (i.e., central low echo) of BUS view.

2. Related work

In this section, we first review previous work on machine learning based CEUS analysis. Then, we briefly introduce the explicit and implicit data augmentation methods for enriching sample diversity.

2.1. Machine learning based CEUS analysis

Previous studies on machine learning-based dual-view CEUS analysis can be broadly classified into two categories: (1) multi-view learning (Guo et al., 2018; Han et al., 2023; Li et al., 2024), which focuses on the effective fusion of features from BUS and CEUS views, and (2) spatial-temporal feature aggregation (Chen et al., 2021; Liu et al., 2020; Qin et al., 2023), which emphasizes the extraction and aggregation of dynamic perfusion features. Guo et al. (2018) developed a multi-view classification framework for liver CEUS, integrating enhancement patterns from the arterial, portal venous, and delayed phases to differentiate between benign and malignant lesions. Subsequent research primarily focused on transferring complementary CEUS information to the BUS view for a more comprehensive evaluation (Han et al., 2023; Zhang et al., 2023a). Another line of work emphasizes dynamic perfusion feature learning utilizing spatial-temporal convolution or temporal attention mechanisms. Zhao et al. (2023) combined convolution neural networks with bidirectional long short-term memory networks to predict the early recurrence probability of liver metastases. Chen et al. (2021) employed clinical attention priors to guide the network's focus on critical points. Although these methods have demonstrated promising results in dual-view CEUS classification, they still depend on collecting and annotating of sufficient CEUS data, which is often challenging in clinic where data are scarce.

2.2. Explicit and implicit data augmentation

Data augmentation has been the prevalent technique to reduce the risk of overfitting common in machine learning (Shankar et al., 2018; Volpi et al., 2018; Su et al., 2023; Zhang et al., 2019; Huang et al., 2018). By generating artificial data, machine learning models are expected to capture causal correlation between attributes and class with the boosted sample diversity. The most common image augmentation method is random transformation, like flipping, cropping, and color shifting, etc. Despite their simplicity, these methods are inherently limited as they operate in an unsupervised manner, introducing variations that do not necessarily align with meaningful semantic changes in the data. To introduce more targeted semantic variations, image-to-image translation framework is widely used to augment images with rich semantic modifications (Huang et al., 2018). To alter the specific aspect of semantic, these methods decompose images into variables of interest and achieve controllable image generation. However, image generative models (VAE Doersch, 2016, GAN Karras

Table 1
Notations and descriptions.

Notation	Description
$D = \{(\mathbf{I}_i^{us}, \mathbf{I}_i^{ceus}), y_i\}_{i=1}^N$	Training set of N dual-view CEUS videos.
$\mathbf{f}_i = [f_i^{us}; f_i^{ceus}]$	Original dual-view representation extracted by the backbone networks $\mathcal{F}_\theta^{us}$ and $\mathcal{F}_\theta^{ceus}$.
$\tilde{\mathbf{f}}_i = [\tilde{f}_i^{us}; \tilde{f}_i^{ceus}]$	Augmented dual-view representation obtained via the collaborative semantic reasoning operator $C(\cdot)$.
$f_i^{ceus} = [f_p^{ceus}; \gamma]$	CEUS view representation, composed of the static enhancement texture f_p^{ceus} at the peak and the dynamic enhancement pattern γ .
$\gamma = \mathcal{R}(\mathbf{F}^{ceus})$	Dynamic enhancement pattern derived via temporal rank pooling.
$\mathbf{F}^{ceus}$	Sequence of enhancement texture representations extracted over time from the CEUS view.

et al., 2021b, Diffusion model Gong et al., 2023) are challenging to train. Thus, we turn to the implicit data augmentation strategy at the cost of negligible computational overhead (Wang et al., 2019; Li et al., 2021; Verma et al., 2019). Leveraging the property that deep feature space is well linearized (Upchurch et al., 2017), shifting deep feature along certain direction could alter one aspect of semantic (Li et al., 2021) while preserve its class identity. Additionally, multi-view learning enhances feature representation by integrating complementary information from multiple perspectives. Luo et al. (2023) employs a homogeneous bilinear framework to augment data with multi-resolution tumor-centered images (TCIs), capturing both tumor and surrounding tissue features. Inspired by this, our work extends multi-view augmentation by ensuring inter-modality consistency, improving feature alignment in dual-view ultrasound imaging.

However, most existing studies are designed for single-modality setting without taking inter-modality consistency into account. In this work, we explore an implicit augmentation strategy that preserves inter-modal coherence while improving feature robustness in a multi-view setting.

3. Preliminary

3.1. Notations

Let $D = \{(\mathbf{I}_i^{us}, \mathbf{I}_i^{ceus}), y_i\}_{i=1}^N$ be the training set consisting of N dual-view CEUS videos, where y_i is the class label of the i_{th} dual-view CEUS sample, composed of the BUS view $\mathbf{I}_i^{us}$ and the CEUS view $\mathbf{I}_i^{ceus}$. The original dual-view CEUS representation is denoted by $\mathbf{f}_i = [f_i^{us}; f_i^{ceus}] \in \mathbb{R}^{d_1+3 \times d_2}$ extracted by the backbone network $\mathcal{F}_\theta^{us}$ and $\mathcal{F}_\theta^{ceus}$ of each view. Its augmented version is denoted as $\tilde{\mathbf{f}}_i = [\tilde{f}_i^{us}; \tilde{f}_i^{ceus}] = C(\mathbf{f}_i)$, where $C(\cdot)$ is the operator of collaborative semantics reasoning. Here, CEUS view representation $f_i^{ceus} = [f_p^{ceus}; \gamma] \in \mathbb{R}^{3 \times d_2}$ comprises the static enhancement texture $f_p^{ceus} \in \mathbb{R}^{d_2}$ at the peak point p , and the dynamic enhancement pattern γ obtained by temporal rank pooling $\gamma = \mathcal{R}(\mathbf{F}^{ceus}) \in \mathbb{R}^{2 \times d_2}$, where $\mathbf{F}^{ceus} \in \mathbb{R}^{d_2 \times T}$ is the sequence of enhancement texture representations, with the dimensionality of d_2 and temporal length of T . For simplicity, we omit the subscript i denoting the i_{th} sample (see Table 1).

3.2. Implicit data augmentation

Implicit data augmentation (IDA) refers to generating artificial data by operating in feature space rather than directly tweaking the input data. By translating original samples along specific directions that correspond to meaningful semantic changes, IDA allows for a much more efficient way to enrich sample semantics. This is particularly beneficial in our dual-view CEUS classification where data is scarce and expensive to collect.

Given an input sample feature $\mathbf{f}_i \in \mathbb{R}^d$ from the class y_i , the augmented feature $\tilde{\mathbf{f}}_i$ is produced by shifting features towards specific semantic direction $\mathbf{v}_i$, sampled from the augmentation distribution $\mathcal{N}(\mathbf{0}, \Sigma_i)$,

$$\tilde{\mathbf{f}}_i = \mathbf{f}_i + \alpha \mathbf{v}_i \quad (1)$$

$$\mathbf{v}_i \sim \mathcal{N}(\mathbf{0}, \Sigma_i) \quad (2)$$

where α is a scaling factor that controls the strength of semantic perturbation. $\mathcal{N}(\mathbf{0}, \Sigma_i)$ is a normal distribution with zero mean $\mathbf{0}$ and the covariance matrix Σ_i capturing the semantic variance around $\mathbf{f}_i$.

In previous work, samples within the same class typically share an augmentation distribution Σ_i specific to the class y_i , which overlooks the variability among individual samples within that class. However, dual-view CEUS images often exhibit large intra-class variance, for example, significant texture differences could be observed even within the same type of liver cancer. More importantly, existing IDA methods are primarily designed for single-modality settings, where independently augmented views may not be semantically aligned. Based on this, we propose to perform a sample-adaptive data augmentation with collaborative semantic reasoning across views.

4. Method

In this part, we introduce in detail our proposed CRIDA method, including the architecture of our network (Section 4.1), dual-view CEUS representation (Section 4.2), the collaborative semantics reasoning strategy (Section 4.3), a modified cross-entropy loss for model training (Section 4.4), and the implementation details (Section 4.5).

4.1. Architecture

The overall architecture of the proposed Collaborative Reasoning-based Implicit Data Augmentation (CRIDA) method is depicted in Fig. 2. It consists of two sequential components, i.e., (1) raw dual-view CEUS representation learning and (2) collaborative reasoning-based semantic augmentation.

Our CRIDA method first extracts BUS view representation f_i^{us} and sequential CEUS view representation $\mathbf{F}_i^{ceus}$ through the backbone network $\mathcal{F}_\theta^{us}$ and $\mathcal{F}_\theta^{ceus}$, respectively. Then, dynamic enhancement pattern γ_i is obtained by temporal ranking pooling $\gamma = \mathcal{R}(\mathbf{F}_i^{ceus})$, concatenated with the peaking enhancement pattern $f_{i,p}^{ceus}$ to represent the dynamic blood perfusion process. After that, dual-view CEUS features $\mathbf{f}_i = [f_i^{us}; f_i^{ceus}]$ are augmented by the module of collaborative semantic reasoning $C(\cdot)$, which transfers relevant semantics via cross-view collaborative reasoning. Finally, augmented dual-view CEUS features $\tilde{\mathbf{f}}_i = [\tilde{f}_i^{us}; \tilde{f}_i^{ceus}]$ are concatenated for lesion classification.

4.2. Dual-view CEUS representation

Our proposed CRIDA method adopts dual-view CEUS imaging as the inputs, which consists of static BUS view and dynamic CEUS view. Since the fact that dynamic enhancement patterns, such as the orientation and velocity of contrast enhancement or diffusion, hold significant diagnostic value for lesion evaluation, we first need to encode dynamic enhancement pattern in CEUS view.

Given the sequence of enhancement texture representation $\mathbf{F}^{ceus}$, we adopt the temporal rank pooling (TRP) (Fernando et al., 2016; Cherian et al., 2017) to capture dynamic enhancement pattern, $\gamma = \mathcal{R}(\mathbf{F}^{ceus})$. The rationale behind using TRP is that it seeks an explicit representation of motion information in video data by assuming that successive video frames evolve along a certain direction in the latent space. In the context of CEUS view representation, this direction could be used to represent dynamic evolution of contrast enhancement textures.

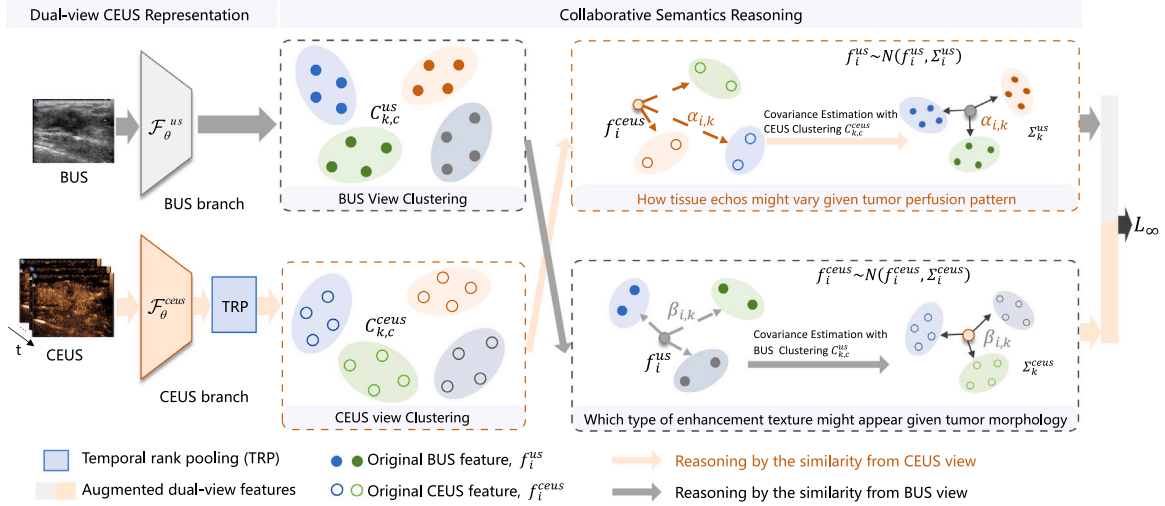


Fig. 2. The framework of the proposed Collaborative Reasoning-based Implicit Data Augmentation (CRIDA) method. It consists of Dual-view CEUS representation and Collaborative semantics reasoning. To ensure augmented BUS and CEUS features to be aligned semantically, this method augments dual-view samples through cross-view collaborative reasoning, addressing the question *how tissue echos might vary given tumor perfusion pattern* and *which type of enhancement texture might appear given tumor morphology*.

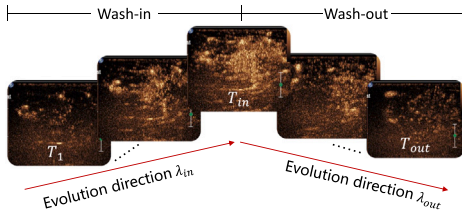


Fig. 3. Dynamic enhancement pattern representation. There exists a temporal evolution direction within wash-in ($T_1 \rightarrow T_{in}$) or wash-out ($T_{in} \rightarrow T_{out}$) stage, γ_{in} and γ_{out} , outlining enhancement texture changes along time.

We observe that a complete perfusion process consists of two distinct stages: wash-in and wash-out phases, split by the point of perfusion peak. As illustrated in Fig. 3, the two stages exhibit opposite trends in blood supply change, transitioning from gradual enhancement to eventual fading. Certainly, dynamic enhancement patterns of the two stages point to distinct evolution directions, prompting us to extract them separately,

$$\gamma_{in} = \mathcal{R}(\mathbf{F}_{in}^{ceus}) \quad (3)$$

$$\gamma_{out} = \mathcal{R}(\mathbf{F}_{out}^{ceus}) \quad (4)$$

where γ_{in} and γ_{out} are dynamic enhancement patterns of the wash-in and wash-out stage, respectively, extracted from the corresponding sequence $\mathbf{F}_{in}^{ceus}$ and $\mathbf{F}_{out}^{ceus}$. We concatenate them to form a complete enhancement pattern representation, $\gamma = [\gamma_{in}; \gamma_{out}] \in \mathbb{R}^{2 \times d_2}$. Finally, the CEUS view is represented as $f_i^{ceus} = [f_p^{ceus}; \gamma] \in \mathbb{R}^{3 \times d_2}$, where $f_p^{ceus} \in \mathbb{R}^{d_2}$ captures the enhancement texture at the peak point p . In this work, we follow an efficient implementation of TRP $\mathcal{R}(\cdot)$ in Bilen et al. (2016), which reduces the original convex optimization to a weighted combination of elements of $\mathbf{F}_{in}^{ceus}$ or $\mathbf{F}_{out}^{ceus}$. Detailed descriptions can be found in Supplementary file.

4.3. Collaborative semantics reasoning

In this section, we elaborate on the process of collaborative semantic reasoning $\mathcal{C}(\cdot)$. With the constructed knowledge graph, we can transfer relevant semantics via cross-view collaborative reasoning.

4.3.1. Sample-wise augmentation distribution

As introduced in Section 3.2, implicit data augmentation is achieved by translating features along certain semantic direction $\mathbf{v}$. For example, shifting CEUS view feature f_i^{ceus} along the direction of homogeneous enhancement $\mathbf{v}_{ho}$ might alter the degree of uniform diffusion. In our CRIDA method, semantic direction $\mathbf{v}$ is sampled from a sample-wise augmentation distribution $\mathcal{N}(0, \Sigma_i)$, a Gaussian distribution with $\mathbf{0}$ mean and Σ_i describing the plausible semantic variances of i_{th} sample. In this way, augmented feature $\tilde{f}_i = f_i + \mathbf{v}_i$ follows the Gaussian distribution,

$$\begin{aligned} \tilde{f}_i &\sim \mathcal{N}(f_i, \Sigma_i) \\ \Sigma_i &= \begin{bmatrix} \Sigma_i^{us} & \\ & \Sigma_i^{ceus} \end{bmatrix} \end{aligned} \quad (5)$$

where Σ_i is a block diagonal matrix composed of $\Sigma_i^{us} \in \mathbb{R}^{d_1 \times d_1}$ and $\Sigma_i^{ceus} \in \mathbb{R}^{3d_2 \times 3d_2}$. It is noted that covariance matrix Σ_i is estimated for each sample, which differs from traditional augmentation distributions estimated at the class level (Wang et al., 2019; Hong et al., 2022), which overlooks that samples in the same class might be affected differently with respect to distinct attributes.

4.3.2. Knowledge graph construction

To achieve sample-wise data augmentation with collaborative reasoning, we construct a sample-wise knowledge graph $\mathcal{G}_i$ defining sample-to-cluster semantic similarity. For each view, the nodes of $\mathcal{G}_i$ consist of the i th sample and K cluster prototypes, while the edges measure the semantic similarity between the sample and clusters for semantic transfer.

Specifically, we first train the backbone network without augmentation for few epochs, and then group training samples $\mathcal{D}$ into K clusters independently for each view within each class,

$$\mathcal{C}_{k,c}^{us} = \text{Clust}(\{f_i^{us} | y_i = c\}) \quad (6)$$

$$\mathcal{C}_{k,c}^{ceus} = \text{Clust}(\{f_i^{ceus} | y_i = c\}) \quad (7)$$

where $\mathcal{C}_{k,c}^{us}/\mathcal{C}_{k,c}^{ceus}$ denotes the k_{th} cluster among BUS/CEUS view belong to the class c . Within each cluster, instances have the similar morphological/functional features and the same diagnostic label $y_i = c$.

For each view, edges of knowledge graph $\mathcal{G}_i$ are calculated as,

$$\alpha_{i,k} = \frac{1}{\alpha_i} \exp\left(-\frac{1}{2s^2} \|f_i^{ceus} - f_k^{ceus}\|_2\right) \times N_k^{ceus} \quad (8)$$

$$\beta_{i,k} = \frac{1}{\beta_i} \exp \left(-\frac{1}{2s^2} \|f_i^{us} - f_k^{us}\|_2 \right) \times N_k^{us} \quad (9)$$

where $\alpha_{i,k}/\beta_{i,k}$ measures the similarity between i_{th} sample and k_{th} cluster in CEUS/BUS view. N_k^{us}/N_k^{ceus} is the k_{th} cluster size and f_k^{ceus}/f_k^{us} is the cluster prototype. It is noted that the strength of transferred semantic variance is not only determined by the similarity between sample features and cluster prototype, but also by the size of associated cluster N_k^{us} (N_k^{ceus}), since the larger cluster could contain the more accurate variance matrix. α_i and β_i are the normalized coefficient, $\alpha_i = \sum_k \alpha_{i,k}$ and $\beta_i = \sum_k \beta_{i,k}$. Gaussian parameter s adjusts the flexibility of semantic association. Note that, similarity calculation is restricted within the same class to keep the class identity. For simplicity, we omit the subscript c denoting the class.

4.4.3. Collaborative reasoning-based augmentation

As mentioned, the semantically augmented BUS and CEUS view features $\tilde{f}_i^{ceus}$ and $\tilde{f}_i^{ceus}$ are expected to be aligned in the feature space. When projected back into the imaging space, they are likely derived from the same lesion section.

To address the question *how tissue echos might vary given tumor perfusion pattern*, we explore the transferable echo changes from similar clusters in the CEUS view, propagating echo variations from instances exhibiting similar enhancement patterns. Thus, sample-wise covariance matrix of BUS view Σ_i^{us} in Eq. (5) is calculated as,

$$\Sigma_i^{us} = \sum_{k=1}^K \alpha_{i,k} \Sigma_k^{us} \quad (10)$$

where $\alpha_{i,k}$ is the sample-to-cluster similarity score defined in Eq. (8), used to measure the amount of transferred semantic information from different clusters. Note that, Σ_k^{us} is estimated using BUS view features f_i^{us} from the k_{th} cluster in CEUS view.

$$\Sigma_k^{us} = \text{Var}(\{f_i^{us}\}), i \in C_{k,c}^{ceus} \quad (11)$$

With the guidance of cross-view clustering information, we could reason the plausible echo changes under the same type of enhancement pattern. Similarly, to answer the question *which type of enhancement texture might appear given tumor morphology*, we could borrow semantic directions from instances with approximate tumor morphology. Thus, sample-wise covariance matrix of CEUS view Σ_i^{ceus} in Eq. (5) is calculated as,

$$\Sigma_i^{ceus} = \sum_{k=1}^K \beta_{i,k} \Sigma_k^{ceus} \quad (12)$$

where $\beta_{i,k}$ is defined as Eq. (9), and Σ_k^{ceus} is estimated using CEUS view features f_i^{ceus} from the k_{th} cluster in BUS view.

$$\Sigma_k^{ceus} = \text{Var}(\{f_i^{ceus}\}), i \in C_{k,c}^{us} \quad (13)$$

4.4.4. Online covariance matrix estimation

In this study, sample grouping is implemented by K means algorithm after the pre-training of backbones, and these group assignments are fixed throughout the subsequent training phase. To efficiently estimate the covariance matrices as defined in Eqs. (11) and (13) are updated in an online manner. Specifically, we employ an online update strategy that continuously incorporates new feature representations as training progresses.

Specifically, we introduce a caching queue, denoted as $\mathbb{Q}$, which serves as a dynamic repository for the deep feature representations extracted from both modalities. As each mini-batch $\mathcal{B}$ is processed, the latest feature representations for each sample are computed and used to update the cache in a batch-by-batch manner. Formally, for each sample i , the features for the BUS and CEUS modalities are updated in the cache as follows:

$$\begin{aligned} f_i^{\mathbb{Q}_{us}} &\leftarrow f_i^{us} \\ f_i^{\mathbb{Q}_{ceus}} &\leftarrow f_i^{ceus} \end{aligned} \quad (14)$$

where i is the sample index within a batch $\mathcal{B}$, $i \in \mathcal{B}$.

4.4. Modified cross-entropy loss

4.4.1. Cross-entropy loss on augmented features

Assuming we have performed semantic augmentation on the i_{th} training sample $f_i = [f_i^{us}, f_i^{ceus}]$ for M times, $\{\tilde{f}_i^1, \tilde{f}_i^2, \dots, \tilde{f}_i^M\}$, we could optimize the model with cross-entropy (CE) loss on the augmented features $\tilde{f}_i$ as follows,

$$\mathcal{L}_M(\mathbf{W}, \mathbf{b}, \vartheta) = \frac{1}{NM} \sum_{i=1}^N \sum_{m=1}^M -\log \frac{\exp(\mathbf{w}_{y_i}^T \tilde{f}_i^m + b_{y_i})}{\sum_c \exp(\mathbf{w}_c^T \tilde{f}_i^m + b_c)} \quad (15)$$

where $\mathbf{W}$ and $\mathbf{b}$ are the weight matrix and bias vector of the last fully connected layer, respectively. ϑ denotes the set of remaining network parameters. When M is large, Eq. (15) becomes computationally cost to implement. Thus, we derive the upper bound of this loss function $\mathcal{L}_\infty$, which allows for a much more efficient computation.

4.4.2. Upper bound of the loss function

As shown in Eq. (5), augmented samples follow Gaussian distribution. We could derive the expectation of Eq. (15) when M approaches infinity. According to Jensen's inequality $E[\log(X)] \leq \log(E[X])$, its upper bound is obtained by,

$$\begin{aligned} \mathcal{L}_\infty(\mathbf{W}, \mathbf{b}, \vartheta) &= \frac{1}{C_\infty} \sum_{i=1}^N E_{\tilde{f}_i} \left[-\log \frac{\exp(\mathbf{w}_{y_i}^T \tilde{f}_i + b_{y_i})}{\sum_c \exp(\mathbf{w}_c^T \tilde{f}_i + b_c)} \right] \\ &\leq \frac{1}{C_\infty} \sum_{i=1}^N \log \left(E_{\tilde{f}_i} \left[\sum_c \exp(A_i^c) \right] \right) \\ &= \frac{1}{C_\infty} \sum_{i=1}^N \log \left(\sum_{c=1}^C E_{\tilde{f}_i} [\exp(A_i^c)] \right) \end{aligned} \quad (16)$$

where C_∞ is the normalized coefficient. $A_i^c = (\mathbf{w}_c^T - \mathbf{w}_{y_i}^T) \tilde{f}_i + (b_c - b_{y_i})$ is random variable following Gaussian distribution $\mathcal{N}(\mu_i^c, \sigma_i^c)$. Leveraging the property of Gaussian distribution, $E[\exp(mX)] = \exp(m\mu + \frac{1}{2}m^2\sigma)$, $X \sim \mathcal{N}(\mu, \sigma)$, we could derive the upper bound as,

$$\mathcal{L}_\infty = \frac{1}{N} \sum_{i=1}^N \log \left(\sum_{c=1}^C \exp \left(\mu_i^c + \frac{1}{2} \sigma_i^c \right) \right) \quad (17)$$

where $\mu_i^c = (\mathbf{w}_c - \mathbf{w}_{y_i})^T f_i + (b_c - b_{y_i})$ and $\sigma_i^c = (\mathbf{w}_c - \mathbf{w}_{y_i})^T \Sigma_i (\mathbf{w}_c - \mathbf{w}_{y_i})$ based on Eq. (5).

4.4.3. Cross-entropy like loss

Alternatively, we could reformulate the upper bound as the cross-entropy like loss,

$$\begin{aligned} \mathcal{L}_\infty &= \frac{1}{N} \sum_{i=1}^N -\log \left(\frac{\exp(\mathbf{w}_{y_i}^T f_i + b_{y_i})}{\sum_c \exp(\mathbf{w}_c^T f_i + b_c + \frac{1}{2} \sigma_i^c)} \right) \\ &= \frac{1}{N} \sum_{i=1}^N -\log \left(\frac{\exp(Z_i^{y_i})}{\sum_c \exp(Z_i^c)} \right) \end{aligned} \quad (18)$$

where $Z_i^c = z_i^c + \frac{1}{2} \sigma_i^c$ composed of the logit output of the class c , $z_i^c = \mathbf{w}_c^T f_i + b_c$ and bias σ_i^c . Note that, $\sigma_i^{y_i}$ equals to zero as defined. As observed, when M grows to infinity, the derived upper bound $\mathcal{L}_\infty$ reduces to a standard CE loss that allows for a much more efficient computation.

4.5. Implementation details

In this study, we implement the proposed Collaborative Reasoning-based Implicit Data Augmentation (CRIDA) method using Pytorch 1.8.1 and run on a single GPU (NVIDIA RTX 3090, 24 GB). Our method is

Algorithm 1 CRIDA Method.

Input: A training set of dual-view CEUS sequences $\mathcal{D} = \{(\mathbf{f}_i^{us}, \mathbf{f}_i^{ceus}), \mathbf{y}_i\}_{i=1}^N$

Output: Trained CRIDA model with backbones $\mathcal{F}_\theta^{us}$, $\mathcal{F}_\theta^{ceus}$, and prediction head $\mathbf{W}$, $\mathbf{b}$

- 1: **Knowledge Graph Construction:**
- 2: *Initial Training:* Train the network for a few epochs using the raw deep representation. $\triangleright$ Eq.(3)-(4)
- 3: *Sample Clustering:* Extract dual-view CEUS representations $\mathbf{f}_i^{us}$ and $\mathbf{f}_i^{ceus}$. Cluster the training samples $\mathcal{D}$ into K clusters for each view within each class, $\mathcal{C}_{k,c}^{us}$, $\mathcal{C}_{k,c}^{ceus}$
- 4: *Sample-to-cluster Similarity:* Initialize update covariance matrices Σ_k^{us} and Σ_k^{ceus} . Construct the sample-wise knowledge graph $\mathcal{G}_i$. $\triangleright$ Eq.(8), (9)
- 5: **Collaborative Reasoning-based Augmentation:**
- 6: Estimate the covariance matrices through collaborative reasoning using knowledge graph $\mathcal{G}_i$. $\triangleright$ Eq.(10), (12)
- 7: **while** $\mathcal{L}_\infty$ has not converged **do**
- 8: **for** each mini-batch $B \subset \mathcal{D}$ **do**
- 9: Update network parameters using the Adam optimizer. $\triangleright$ Eq.(18)
- 10: Update the caching queue $\mathbb{Q}$ incrementally. $\triangleright$ Eq.(14)
- 11: Augment dual-view features $\mathbf{f}_i = [\mathbf{f}_i^{us}, \mathbf{f}_i^{ceus}]$ through collaborative semantic reasoning by repeating Step 4.
- 12: **end for**
- 13: **end while**

constructed on a dual-branch network that separates feature extraction of each modality with Resnet-18 as backbone, which was selected for its balance between computational efficiency and feature representation. Additionally, we leveraged ImageNet pre-trained weights to enhance feature extraction and improve generalization. We optimize the network with Adam optimizer with the learning rate of 0.0003. To obtain the initial clustering of samples, we first pre-train the backbone for $N_{pre} = 20$ epochs. After that, we fix the sample grouping and train the model with the modified cross-entropy loss (Eq. (4.4)) with the maximal 50 epochs. The dimension of $\mathbf{f}^{us}$ is set to $d_1 = 512$ and $\mathbf{f}^{ceus}$ is $3d_2 = 512 \times 3$. To control the strength of semantic feature augmentation, we add a weighting coefficient λ on the added item σ_i^c , $\lambda \sigma_i^c$ and increment it in a cosine fashion. λ is chosen from [0.1, 0.25, 0.5, 1.0]. The number of cluster K and Gaussian parameter s^2 are selected from [2, 3, 4, 5] and [0.5, 1, 3, 5, 10], respectively. The optimal hyper-parameter values are determined on the validation performance.

4.6. Complexity analysis

It is noted that the collaborative semantics reasoning components are inactive during inference, ensuring no added runtime cost in deployment. Once the model is trained, our model operates with the same computational footprint as a standard baseline model. The dual-branch architecture, built upon ResNet-18, processes a single BUS image and a sequence of CEUS frames, resulting in a total computational complexity of 29 GFLOPS per subject. The inference time remains highly efficient at 4.6 ms per subject, ensuring real-time performance. For resource-constrained environments, the model can be further optimized by utilizing lighter backbones without compromising performance.

5. Experimental settings

5.1. Dataset

In this study, we collected two groups of dual-view CEUS datasets for model evaluation, including (1) benign/malignant differentiation of breast lesion and (2) sub-type differentiation of primary liver cancer,

Table 2

Parameters of the collected dual-view CEUS datasets. Dataset I and II are breast and liver dual-view CEUS dataset, respectively. 1 denotes malignant lesion or ICC. FPS are the abbreviations for frames per second.

Datasets	No.	Ages	Class(0/1)	FPS	Resolution (pix.)
I	333	49.05 $\pm$ 11.42	114/219	9	800 $\times$ 600
II	202	53.05 $\pm$ 10.03	113/89	9	800 $\times$ 600

i.e., Hepatocellular carcinoma (HCC) and Intrahepatic cholangiocarcinoma (ICC).

Both datasets were collected from patients who attended Nanjing Drum Tower Hospital for ultrasound examination from September 2018 to December 2021. All cases have undergone histopathological confirmation. Routine ultrasound examinations were performed on a Philips iU22 scanner (Philips, Bothell, WA) and contrast-enhanced US was performed with the second-generation contrast agents SonoVue (Bracco SpA, Milan, Italy) at a low mechanical index. To ensure imaging quality, each examination was performed by expert radiologists with at least 3-year manipulation experience. During examination, the optimal longitudinal section of the lesion was determined along with BUS view. Radiologists were required to keep the transducer fixed and patients are guided to breathe steadily. Each dual-view CEUS video lasts for about 3 min to record the complete blood perfusion process, starting from the administration of contrast agents and ending until the eventual fading out. Approval was obtained by the Ethics Review Committee of Nanjing Drum Tower Hospital. Detailed information on the two groups of datasets are listed in Table 2.

5.2. Pre-processing

5.2.1. Motion compensation

Respiration motion is the primary interference for contrast-enhanced US imaging, arising from the patient's heartbeat and breathing. To compensate irregular motion interference, we first choose the compact and real-time descriptors (CARD) (Ambai and Yoshida, 2011) to estimate affine translations for the alignment of successive CEUS frames. Furtherly, we extract time-intensity curves (TICs) within lesions regions and remove those points with irregular intensity fluctuations.

The number of points removed from the TIC depends on the extent of irregular fluctuations observed, which in turn varies with imaging acquisition conditions (e.g., free-breathing, cardiac motion, hand-held probe jitter, etc.). Typically, we remove outlier points that deviate beyond a predefined threshold, which generally accounts for about 5%–10% of the total time points. We provide several examples in the Supplementary file for clarity.

5.2.2. Temporal pruning

Contrary to the high temporal resolution of CEUS videos, the changes of enhancement texture tend to be smoothing and slow. To filter redundant CEUS frames, we preserve one single CEUS frame per second following previous studies (Feng et al., 2018; Liu et al., 2020) which is sufficient to extract enhancement dynamics using rank pooling operation. We measured an average inference time of 4.6 ms at 1 fps, demonstrating its feasibility for real-time applications.

5.3. Evaluation protocol

For model evaluation, we performed a random split of the entire dataset by preserving the class ratio, allocating 80% for training and 20% for testing. Among which, 25% of training data is held out for model validation to determine the optimal hyper-parameter values. As random partition is involved, we report the mean and standard deviation (std.) of the testing results for 5 independent runs. In our experiments, malignant breast lesion and intrahepatic cholangiocarcinoma (ICC) are labeled as the positive class due to the higher diagnostic value in clinics. The classification performance was evaluated by five metrics, including Area under curve (AUC), classification accuracy (ACC), sensitivity (SEN), specificity (SPE), and F1-score.

Table 3

Diagnostic performances comparison of various methods. The terms a and b in “a ± b” denote the mean and standard deviation (std.).

Method	AUC	ACC	SEN	SPE	F1-score
Differentiation of benign and malignant breast lesion					
DCCA	83.20 ± 3.66	83.58 ± 1.06	90.45 ± 1.90	70.43 ± 3.64	87.86 ± 0.81
DCCAE	82.51 ± 4.20	81.19 ± 1.70	85.91 ± 4.07	72.17 ± 6.59	85.69 ± 1.52
MVCNN	83.60 ± 1.96	83.58 ± 1.06	90.91 ± 4.55	69.57 ± 7.53	87.89 ± 1.07
MM Dynamics	84.92 ± 5.64	86.27 ± 1.95	90.10 ± 2.27	71.04 ± 5.67	87.92 ± 1.39
TMC	84.46 ± 2.97	84.87 ± 2.71	92.18 ± 3.59	72.77 ± 5.53	88.93 ± 1.90
QMF	86.11 ± 2.34	85.97 ± 2.26	93.18 ± 2.78	73.17 ± 2.38	89.70 ± 1.77
RCML	88.46 ± 4.93	85.67 ± 2.71	92.73 ± 5.18	72.17 ± 9.02	89.45 ± 2.19
CRIDA (Ours)	90.14 ± 2.08	89.25 ± 0.67	93.64 ± 2.49	80.87 ± 3.89	91.96 ± 0.61
Differentiation of primary liver cancer					
DCCA	86.75 ± 3.39	81.25 ± 4.94	81.90 ± 1.37	80.00 ± 9.96	84.84 ± 5.14
DCCAE	92.29 ± 6.79	85.00 ± 5.59	85.71 ± 9.52	83.64 ± 7.61	88.05 ± 4.88
MVCNN	88.91 ± 1.63	87.08 ± 0.93	74.67 ± 5.77	96.30 ± 5.24	83.19 ± 1.09
MM Dynamics	94.41 ± 2.80	91.25 ± 1.74	86.43 ± 3.93	92.89 ± 3.29	88.45 ± 2.07
TMC	91.31 ± 4.39	89.06 ± 2.62	81.96 ± 7.03	94.51 ± 2.06	86.52 ± 3.82
QMF	94.42 ± 4.55	91.67 ± 3.29	88.14 ± 9.03	94.21 ± 4.20	89.80 ± 4.07
RCML	93.64 ± 1.36	89.58 ± 1.47	91.14 ± 4.22	88.39 ± 3.10	88.13 ± 1.73
CRIDA (Ours)	96.00 ± 1.86	95.57 ± 1.64	92.00 ± 4.47	97.14 ± 3.91	93.89 ± 2.25

5.4. Competing methods

To investigate the improvement of the proposed CRIDA method, we compare it with several state-of-the-art multi-view learning methods as follows:

DCCA (Andrew et al., 2013): Deep Canonical Correlation Analysis extends the traditional linear Canonical Correlation Analysis (CCA) into deep neural networks, enabling it to maximize correlations and capture complex, nonlinear relationships between multiple views.

DCCAE (Wang et al., 2015): Deep Canonically Correlated Auto-Encoder builds on DCCA by incorporating an auto-encoder framework, effectively maximizing inter-view correlations while preserving view-specific information to ensure richer feature representations.

MVCNN (Su et al., 2015): Multi-view Convolution Neural Network aggregates view-specific features into a global representation, leveraging both individual and complementary information from multiple views.

MM Dynamics (Han et al., 2022a): Multi-modal Dynamics evaluates the informativeness of both feature-level and modality-level signals for different samples, guiding a more reliable and trustworthy fusion process.

TMC (Han et al., 2022b): Trusted Multi-View Classification dynamically assesses the quality of different views and integrates them at the evidence level using a Dirichlet distribution.

QMF (Zhang et al., 2023b) Quality-aware Multimodal Fusion addresses the challenge of noisy multi-view inputs by dynamically adjusting the fusion process, prioritizing the most reliable information from each modality.

RCML (Xu et al., 2024) Reliable Conflictive Multiview Learning tackles the issue of conflicting views by calculating the degree of conflict and conjunctive certainty among views, ensuring robust and reliable multi-view fusion.

Notably, we adopted the same pre-processing steps described in Section 5.2 for these competing methods. For multi-view classification, gray-scale US, CEUS frame at the peak point and the global enhancement dynamics described in Section 4.2 are adopted as three input views.

6. Results

In this section, we first compare our CRIDA method with several state-of-the-art methods. Then, we validate the effectiveness of dynamic enhancement pattern representation and collaborative semantics reasoning of this method. After that, we perform ablation experiments by removing the core components. Finally, we analyze the sensitivity of key model hyper-parameters and provide visualization results of augmented samples.

6.1. Comparison on diagnostic performance

In this group of experiments, we compare the diagnostic performance of our CRIDA method with several state-of-the-art multi-view learning methods. Quantitative results on the task of breast lesion and liver cancer differentiation are presented in Table 3.

As observed from the experimental results, our CRIDA method consistently achieves clear performance improvements on both tasks. As shown in Table 3, our CRIDA method and RCML rank the first and second for the task of breast lesion differentiation, hitting AUC score of 90.14% and 88.46%. The associated AUC scores of the remaining methods are below 87%. We note that most competing methods could obtain a high diagnostic sensitivity around 90% but the counterpart specificity scores are much lower. The highest specificity score is obtained by our CRIDA method with 80.87% and other competing methods only achieve specificity around 73%. The significantly increased specificity confirms that there are lower chances of the proposed CRIDA method to misclassify benign lesions as malignant lesions, reducing the risk of over-treatment. The inferior performances are obtained by QMF, TMC and MM Dynamics with AUC around 85%, while the least diagnostic performances are observed with CCA-based methods (i.e., DCCA and DCCAE), hitting AUC score around 83%.

From Table 3, we could see that our CRIDA method achieves the highest ACC of 95.57% and AUC of 96% on the task of liver cancer differentiation, while the accuracy of the remaining methods do not exceed 92%. Similar to Table 3, most of the performance improvements are statistically significant, especially for ACC and F1-score. Different from breast lesion differentiation, the diagnostic sensitivity of liver cancer is much lower than the specificity, which implies that ICC cases are easier to be misclassified as HCC cases. The highest sensitivity score is obtained by our CRIDA method with 92% and other competing methods mostly achieve the sensitivity around 85%.

Comparing the results of two datasets, it is evident that both our CRIDA method and the competing methods achieve higher classification scores in the task of liver cancer diagnosis, indicating that differentiating breast lesions is more challenging using dual-view CEUS imaging. Therefore, we could state that the better results in breast lesion differentiation further validate the superiority of our method with enhancement dynamics learning and collaborative semantic augmentation.

6.2. Comparison of temporal dynamics learning

As introduced in Section 4.2, dynamic enhancement pattern γ_{in} and γ_{out} from wash-in and wash-out stages functions as a key component for dual-view CEUS representation. To evaluate its effectiveness, we

Table 4
Comparison of different temporal dynamics learning methods.

Method	Breast lesion				
	AUC	ACC	SEN	SPE	F1-score
AP	85.15 $\pm$ 5.13	82.39 $\pm$ 3.56	91.82 $\pm$ 4.43	64.35 $\pm$ 7.78	87.24 $\pm$ 2.61
SA	84.31 $\pm$ 2.52	83.88 $\pm$ 1.95	89.55 $\pm$ 2.59	73.04 $\pm$ 7.78	87.96 $\pm$ 1.31
LSSL	87.09 $\pm$ 0.82	85.07 $\pm$ 1.49	88.18 $\pm$ 2.49	79.13 $\pm$ 6.45	88.59 $\pm$ 1.05
TRP-once	88.42 $\pm$ 1.24	86.87 $\pm$ 3.24	91.36 $\pm$ 4.37	78.26 $\pm$ 6.87	90.12 $\pm$ 2.49
CRIDA	90.14 $\pm$ 2.08	89.25 $\pm$ 0.67	93.64 $\pm$ 2.49	80.87 $\pm$ 3.89	91.96 $\pm$ 0.61
Primary liver cancer					
AP	87.38 $\pm$ 1.29	86.67 $\pm$ 2.80	78.81 $\pm$ 5.61	92.59 $\pm$ 5.86	83.70 $\pm$ 3.00
SA	88.61 $\pm$ 4.32	86.67 $\pm$ 4.80	78.71 $\pm$ 9.85	92.65 $\pm$ 2.62	83.36 $\pm$ 6.89
LSSL	90.04 $\pm$ 0.31	85.63 $\pm$ 1.71	86.67 $\pm$ 5.22	83.64 $\pm$ 7.61	88.74 $\pm$ 1.62
TRP-once	90.61 $\pm$ 0.65	90.42 $\pm$ 1.86	79.57 $\pm$ 2.59	98.54 $\pm$ 1.99	87.67 $\pm$ 2.60
CRIDA	96.00 $\pm$ 1.86	95.57 $\pm$ 1.64	92.00 $\pm$ 4.47	97.14 $\pm$ 3.91	93.89 $\pm$ 2.25

compare it with several variants and the prevailing temporal modeling methods, including (1) Average pooling (AP); (2) Self-attention (SA) (Wang et al., 2018); (3) Longitudinal self-supervised learning (LSSL) (Zhao et al., 2021) that disentangles temporal factors from longitudinal data; (4) **TRP-once**, which derives the enhancement representation of the complete CEUS sequence. Quantitative results on the two tasks are presented in Table 4.

From Table 4, we could have the following observations: (1) deriving an explicit representation of enhancement dynamics is effective to dual-view CEUS fusion classification. It can be verified by the clear performance gains obtained by our CRIDA method and its variant TRP-once on both tasks. The sub-optimal results are obtained by the variant TRP-once, and it still outperforms the remaining methods with AUC of 88.42% and 90.61% on the breast lesion and liver cancer diagnostic tasks, respectively. The difference between our CRIDA method and TRP-once is the separation of wash-in and wash-out phase, which have the opposite blood supply changes. The significantly improved classification performance validates the necessity of this slight modification, which allows for characterizing blood perfusion processes more accurately. Surprisingly, longitudinal self-supervised learning method LSSL achieves an approximate or even worse results compared with temporal aggregation methods. Although this method resorts to an explicit representation of temporal dynamics, its performance might be compromised by the pre-computed dynamic representations learned with an extra auto-encoder architecture. (2) In spite of the similar weighted summation of enhancement patterns, our method yields a significantly higher AUC of 90.14% and 95.57% on the two diagnostic tasks compared with average pooling and self-attention aggregation. It is perhaps due to that pure temporal aggregation methods do not have the benefits of temporal evolution modeling in representation space, degrading the model interpretability from the aspect of representation learning as well.

6.3. Comparison of feature augmentation strategy

As introduced in Section 4.3, collaborative semantics reasoning is introduced to augment the diversity of available dual-view CEUS imaging. To evaluate its effectiveness, we compare it with several variants and other feature augmentation methods, including (1) **Interpolation** (Cheung and Yeung, 2020) which creates harder examples by interpolating to the nearest marginal sample of that class; (2) **Extrapolation** (Cheung and Yeung, 2020) which pulls samples apart from the class center; (3) **ISDA** (Wang et al., 2019) which perturbs modality-specific features by adding Gaussian noise with class-wise distributions; (4) **ISDA-cluster** (Wang et al., 2019) which transfers semantic variances from similar clusters without cross-modality guidance, a variant of our CRIDA method. Quantitative results on the two tasks are presented in Table 5. (5) **CRIDA-aug** which trains exclusively with augmented samples, discarding the original samples.

From Table 5, we could have the following observations: (1) augmenting samples with sample-wise or class-wise statistical information

is proved to be superior to explicit feature manipulation, including interpolation and extrapolation. The optimal results of explicit feature manipulation is obtained by feature extrapolation, hitting AUC scores of 85.85% and 87.65% for breast lesion and liver cancer, respectively. In contrast, the naive implicit feature augmentation method ISDA increases the AUC scores to 88.26% and 88.04% for the two diagnostic tasks, let alone our CRIDA method and its variant ISDA-cluster. The main reason could be that implicit feature augmentation methods guarantee higher sample diversity with abundant semantic variances. (2) Sample-level feature augmentation distribution is more effective than the class-level augmentation distribution, which is verified by the increased classification scores obtained by our CRIDA method and its variant ISDA-cluster. For the more challenging breast lesion differentiation task, we observe that ISDA-cluster with semantics transferred from similar clusters increases the ACC score from 82.09% to 86.87%. (3) Estimating feature augmentation distribution with cross-modality guidance is necessary to further performance improvement in our experiments. By considering semantic consistency of augmented features between modalities, our CRIDA method outperform its variant ISDA-cluster by a large margin on both diagnostic tasks.

Additionally, we observe that even without the original data, cross-modality guided augmentation can achieve competitive performance. CRIDA-aug attains an AUC of 88% for breast lesions and 93.78% for liver cancer. For breast lesions, CRIDA-aug records a sensitivity of 83.64%, confirming that the augmented samples effectively preserve the malignant tumor features critical for diagnosis. Similarly, a sensitivity of 85.38% for liver cancer indicates that the augmented data realistically models the perfusion characteristics in ICC cases. Compared to the baseline V1 in Table 7, CRIDA-aug improves the AUC by 9.38% for breast lesions and 12.54% for liver cancer, underscoring its effectiveness in enriching underrepresented variations when original samples are limited. However, there is a performance drop relative to the full CRIDA approach (which combines original and augmented data). For example, breast lesion performance shows a decrease of 2.14% in AUC and 10% in sensitivity. This gap suggests that while CRIDA-aug captures realistic variations, it still struggles to replicate rare edge cases present in the original data, such as exceptionally small tumors or atypical imaging artifacts. Overall, this comprehensive evaluation highlights the value of both implicit augmentation and cross-modality semantic guidance, while also emphasizing the trade-offs when relying solely on augmented data.

6.4. Comparison with single-modality data augmentation classification.

To validate the benefits of multi-view fusion, we compared our dual-view CEUS classification method with a single-modality data augmentation method – Implicit Semantic Data Augmentation for Deep Networks (ISDA) – which performs feature augmentation at the class level (see Table 6).

Table 5
Comparison of different feature augmentation strategies.

Method	Breast lesion				
	AUC	ACC	SEN	SPE	F1-score
Interpolation	84.71 $\pm$ 4.34	83.88 $\pm$ 3.86	89.55 $\pm$ 4.13	73.04 $\pm$ 12.06	87.97 $\pm$ 2.69
Extrapolation	85.85 $\pm$ 1.95	84.18 $\pm$ 4.17	87.73 $\pm$ 4.98	77.39 $\pm$ 6.45	87.89 $\pm$ 3.29
ISDA	88.26 $\pm$ 4.08	82.09 $\pm$ 1.49	89.55 $\pm$ 2.59	67.83 $\pm$ 9.02	86.80 $\pm$ 0.68
ISDA-cluster	87.19 $\pm$ 4.92	86.87 $\pm$ 2.45	94.09 $\pm$ 3.05	73.04 $\pm$ 7.78	90.40 $\pm$ 1.75
CRIDA-aug	88.00 $\pm$ 1.26	80.90 $\pm$ 4.57	83.64 $\pm$ 9.02	75.65 $\pm$ 8.95	84.97 $\pm$ 4.34
CRIDA	90.14 $\pm$ 2.08	89.25 $\pm$ 0.67	93.64 $\pm$ 2.49	80.87 $\pm$ 3.89	91.96 $\pm$ 0.61
Primary liver cancer					
Interpolation	86.74 $\pm$ 8.28	87.50 $\pm$ 2.95	80.95 $\pm$ 15.79	71.85 $\pm$ 16.27	74.39 $\pm$ 7.95
Extrapolation	87.65 $\pm$ 3.37	89.58 $\pm$ 2.55	75.24 $\pm$ 8.52	78.52 $\pm$ 9.59	74.20 $\pm$ 6.75
ISDA	88.04 $\pm$ 4.94	82.08 $\pm$ 2.80	79.05 $\pm$ 10.43	73.33 $\pm$ 9.22	74.04 $\pm$ 6.91
ISDA-cluster	90.90 $\pm$ 5.25	90.83 $\pm$ 1.14	89.52 $\pm$ 7.06	85.93 $\pm$ 5.49	86.20 $\pm$ 3.82
CRIDA-aug	93.78 $\pm$ 1.67	83.75 $\pm$ 3.06	85.38 $\pm$ 5.05	82.67 $\pm$ 8.31	81.82 $\pm$ 2.30
CRIDA	96.00 $\pm$ 1.86	95.57 $\pm$ 1.64	92.00 $\pm$ 4.47	97.14 $\pm$ 3.91	93.89 $\pm$ 2.25

Table 6
Comparison with single-modality data augmentation classification.

Method	Breast lesion				
	AUC	ACC	SEN	SPE	F1-score
ISDA-US	83.00 $\pm$ 3.56	81.49 $\pm$ 2.02	95.00 $\pm$ 2.23	55.65 $\pm$ 8.43	87.10 $\pm$ 1.15
ISDA-CEUS	81.01 $\pm$ 3.96	80.60 $\pm$ 2.50	90.91 $\pm$ 5.93	60.87 $\pm$ 6.15	85.94 $\pm$ 2.38
CRIDA(Ours)	90.14 $\pm$ 2.08	89.25 $\pm$ 0.67	93.64 $\pm$ 2.49	80.87 $\pm$ 3.89	91.96 $\pm$ 0.61
Primary liver cancer					
ISDA-US	88.00 $\pm$ 5.15	85.00 $\pm$ 4.45	81.00 $\pm$ 3.74	87.86 $\pm$ 7.69	82.01 $\pm$ 4.56
ISDA-CEUS	86.00 $\pm$ 1.50	83.75 $\pm$ 2.76	69.00 $\pm$ 4.90	94.29 $\pm$ 4.29	77.95 $\pm$ 3.62
CRIDA(Ours)	96.00 $\pm$ 1.86	95.57 $\pm$ 1.64	92.00 $\pm$ 4.47	97.14 $\pm$ 3.91	93.89 $\pm$ 2.25

Table 7
Results of ablation experiments under different component combinations.

Method	Modules		Breast lesion		Liver cancer	
	Section 4.2	Section 4.3	AUC	ACC	AUC	ACC
V1			78.62 $\pm$ 3.72	81.49 $\pm$ 1.33	81.24 $\pm$ 3.60	84.17 $\pm$ 1.14
V2	✓		80.06 $\pm$ 5.54	82.39 $\pm$ 2.87	82.54 $\pm$ 1.09	84.58 $\pm$ 1.14
V3		✓	82.03 $\pm$ 1.66	81.79 $\pm$ 1.25	84.87 $\pm$ 1.63	85.42 $\pm$ 3.29
Ours	✓	✓	90.14 $\pm$ 2.08	89.25 $\pm$ 0.67	96.00 $\pm$ 1.86	95.57 $\pm$ 1.64

For breast lesion classification, the single-modality approaches achieved AUC scores of 83% (US) and 81.01% (CEUS), with corresponding accuracies of 81.49% and 80.60%. In contrast, our dual-view fusion method achieved an AUC of 90.14% and an accuracy of 89.25%, along with significant improvements in sensitivity, specificity, and F1-score. Similarly, for primary liver cancer, the US-only and CEUS-only methods reported AUCs of 88.00% and 86.00%, respectively, while our approach substantially outperformed both with an AUC of 96.00%. Notably, specificity and overall diagnostic performance were also significantly enhanced with our dual-view strategy. These results underscore the complementary strengths of the two modalities. By combining anatomical details in BUS and perfusion patterns in CEUS, our method achieves a more holistic understanding of the lesion, leading to superior classification performance over the single-view settings.

6.5. Ablation study

To investigate the impact of each component of this method, we conduct a group of ablation experiments in which three variants removes the module of temporal dynamics learning (Section 4.2) or collaborative semantics reasoning (Section 4.3). Quantitative results are recorded in Table 7.

From the ablation experimental results, we could see that our CRIDA method consistently yields better results than three variants. As expected, the sub-optimal results are obtained by V3 that preserves the collaborative feature augmentation module, achieving AUC

of 82.03% and 84.87% on the breast lesion and liver cancer differentiation task, respectively. It further confirms the role of collaborative feature augmentation to improve the performance of limited dual-view CEUS fusion classification. In contrast, on the challenging breast lesion differentiation task, the associated AUC scores drop to around 80% for the variant V1 and V2 that discard the CFA module. Together with results of Table 5, it implies that multi-view feature augmentation with collaborative semantic reasoning is an indispensable step for a stable performance improvement, which has the potential to generalize to more common multi-view learning problems.

6.6. Influence of the weighting coefficient λ

In this group of experiments, we evaluate the influence of weighting coefficient λ on the classification performance. Using liver cancer differentiation as an example, we orderly select $\lambda = \alpha^2$ values from [0, 0.1, 0.25, 0.5, 1.0] and plotted the result curve in Fig. 5.

From Fig. 5, we can observe that the performance of our method initially increases and then slightly decreases as the weighting coefficient increases. When the weighting coefficient rises from 0 to 0.1, the accuracy significantly improves from 85.8% to 92.1%, suggesting that a moderate level of semantic augmentation effectively enriches sample diversity and enhances the model's classification ability. As the weighting coefficient of feature augmentation continues to increase to 0.25, the performance reaches its peak, with an accuracy of 95.6%. However, when λ exceeds 0.25, both AUC and ACC begin to slightly decline. This suggests that excessive semantic augmentation may disrupt semantic consistency across views, leading to feature confusion and negatively

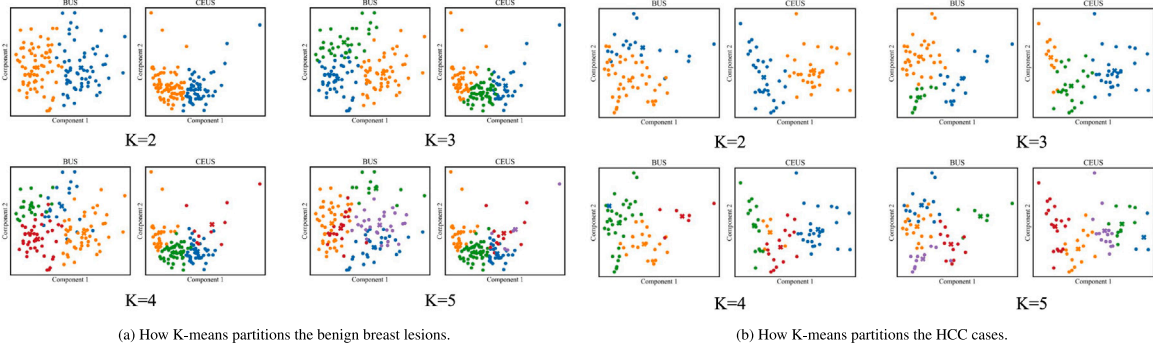


Fig. 4. 2-dim distribution maps of the K -means clustering with various K settings from 2 to 5. Note that, clustering are performed within certain class in our method.

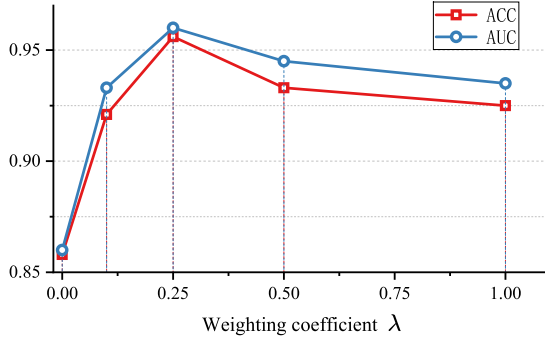


Fig. 5. Results of liver cancer differentiation obtained by our CRIDA method based on different values of the weighting coefficient λ .

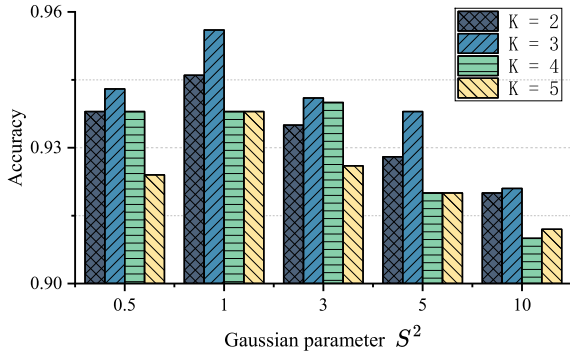


Fig. 6. Results of liver cancer differentiation obtained by our CRIDA method based on different combinations of K and s^2 .

impacting classification performance. Therefore, a moderate level of semantic augmentation strikes the optimal balance, making it the best choice for performance.

6.7. Influence of the number of cluster K and Gaussian parameter s^2

To explore the impact of varying the Gaussian parameter s^2 and the number of clustering categories K on classification performance, we conducted a series of experiments to evaluate the effects of different combinations of these two parameters. Focusing on liver cancer differentiation, we tested values of s^2 from [0.5, 1, 3, 5, 10] and set K to [2, 3, 4, 5], recording the resulting accuracy rates. Additionally, we have included 2-dim distribution maps of the K -means clustering for both BUS and CEUS views in Fig. 4, enabling a further insight into the effect of the parameter K .

As shown in Fig. 6, when s^2 increases from 0.5 to 1, a clear improvement in ACC is observed for all cluster numbers. Specifically,

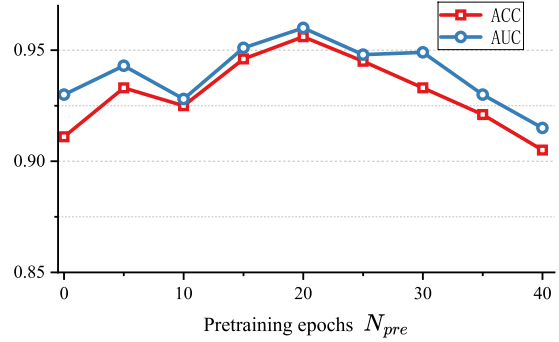


Fig. 7. Results of liver cancer differentiation obtained by our CRIDA method based on different numbers of pretraining epochs N_{pre} .

the highest ACC, near 96%, is achieved with $K = 3$ at $s^2 = 1$. This indicates that a moderate amount of variance, combined with fewer clusters, provides a balance between sufficient data diversity and avoiding excess noise. As s^2 continues to increase beyond 1, we observe a decline in classification accuracy. Particularly, at $s^2 = 5$ and $s^2 = 10$, ACC drops below 93% for all cluster numbers, indicating that larger values of s^2 introduce excessive variance, disrupting the semantic consistency between clusters and reducing the model's classification capability. Regarding the number of clusters K , the optimal performance is consistently observed at $K = 3$. Larger values of K (e.g., $K = 4$ or $K = 5$) generally lead to lower accuracy, implying that increasing the number of clusters beyond 3 adds noise and reduces classification effectiveness. Despite these variations, our method appears to be relatively robust to changes in the number of clusters and variance. The maximum difference in ACC across different s^2 and K combinations is only around 4%, indicating that the method maintains stable performance across a range of parameters.

As observed in Fig. 4, the clustering results for different values of K (ranging from 2 to 5) reveal distinct patterns in both the BUS and CEUS views. When K is set to 2, both views are divided into two broad clusters with a clear separation, indicating that a binary grouping effectively captures the primary structural variations in certain classes. Increasing K to 3 allows for finer separation, potentially segmenting more meaningful subcategories. However, when K is set to larger values (e.g., 4 or 5), the additional clusters further refine the groupings but also introduce the risk of redundancy or noise, which can lead to overfitting.

Overall, the 2-dim distribution maps in Fig. 6 suggest that K values of 2 and 3 are well-suited for our application, striking an optimal balance in capturing the primary structural variations. These findings underscore the importance of carefully selecting the parameter K to ensure that the clustering reflects genuine data structure rather than artifacts. Future work may explore adaptive methods for determining K

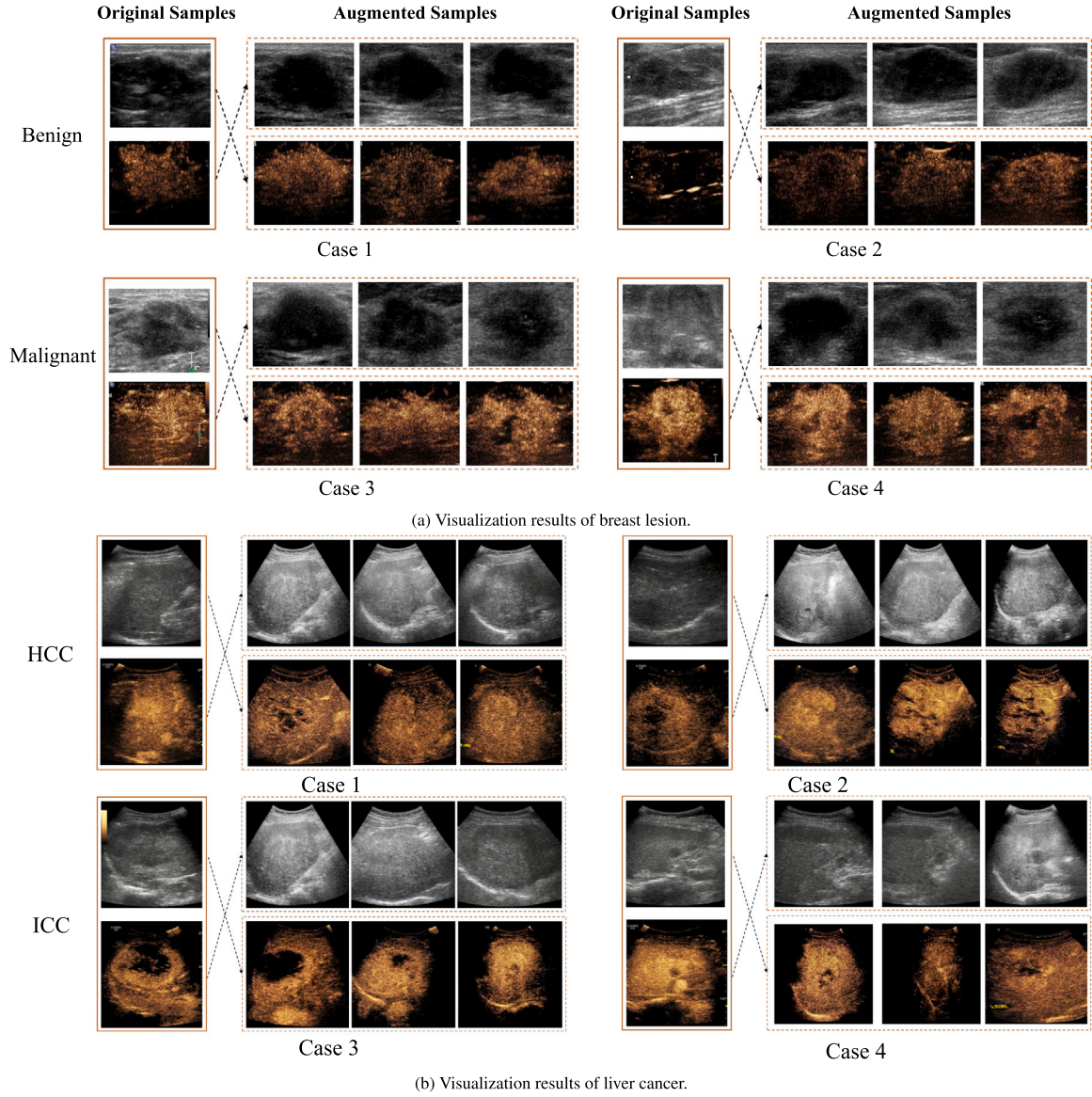


Fig. 8. Visualization of original samples and augmented samples for breast lesion and liver cancer. The dotted line points to the cross-view collaborative reasoning with the given BUS or CEUS view images. For each image, three corresponding augmented images are displayed.

based on the intrinsic properties of the data, thereby further enhancing the robustness and utility of the clustering process.

6.8. Influence of the number of pretraining epochs N_{pre}

In this group of experiments, we investigated the effect of the number of pretraining epochs N_{pre} on the performance of liver cancer differentiation using our CRIDA method. The results are summarized in Fig. 7, where we varied N_{pre} from 0 to 40 and recorded both ACC and AUC.

As shown in Fig. 7, we can see that the overall performance variation is not substantial with the varying pretraining epochs, indicating that pretraining epochs has a relatively minor influence. As N_{pre} increases up to 20 epochs, both the ACC and AUC improve, suggesting that a moderate amount of pretraining helps to better initialize the model, enhancing its ability to cluster features more effectively. However, after 20 epochs, the model's performance begins to gradually decline, and by epoch 40, both ACC and AUC drop significantly, likely due to overfitting and reduced generalization. Nevertheless, it is worth noting that the performance difference across different pretraining

epoch counts is relatively small, with the maximum ACC and AUC variation being around 5%.

6.9. Qualitative analysis

Towards an intuitive understanding of how semantic changes in augmented dual-view CEUS samples, we follow the visualization method proposed in ISDA (Wang et al., 2019) to map augmented features back into the pixel domain. In our experiments, StyleGAN-V3 (Karras et al., 2021a) is trained independently for BUS and CEUS view, allowing for the synthesis of artificial ultrasound images corresponding to augmented samples. We identify the target images in the pixel domain by optimizing the generating factors z within the latent space, synthesizing images that share the same deep representations $\tilde{f}_i^{us}$ and $\tilde{f}_i^{ceus}$.

As shown in Fig. 8, we present the visualization results of two cases for each class of the two tasks. For breast lesion, we could observe that augmented benign samples consistently present a relatively low echo in CEUS view for case 1 and 2. Meanwhile, a higher degree of enhancement with more heterogeneous textures are observed for augmented malignant samples, case 3 and case 4. As for BUS view, we also

find that lesion boundaries of malignant samples appear more blurred compared to those of augmented benign samples. For liver cancer, the augmented BUS view images generally preserve the liver structure while exhibiting subtle variations in texture. In case 4, a round-like lesion with low echogenicity is consistently maintained across the three augmented images. In contrast, the augmented CEUS images reflect variations in the distribution of the contrast agent within the tumor, highlighting distinct vascularization characteristics. For intrahepatic cholangiocarcinoma (ICC), the augmented images show more diffuse or irregular enhancement patterns compared to hepatocellular carcinoma (HCC), consistent with the typical hypovascular nature of ICC tumors. These results suggest that augmented images have the potential to simulate diverse imaging presentations of lesions, which could enhance model generalization in diagnostic applications.

7. Conclusion

In this study, a Collaborative Reasoning-based Implicit Data Augmentation (CRIDA) method is proposed to improve diagnostic performance of dual-view CEUS imaging. To encode dynamic enhancement characteristics of CEUS view, we incorporate temporal rank pooling to disentangle temporal evolution direction of wash-in and wash-out stages, respectively. To boost multi-modality classification performance, we propose to augment dual-view CEUS at the feature level while keep inter-modality consistency. On the collected dual-view CEUS dataset, the proposed method has demonstrated better or at least comparable classification performance.

CRedit authorship contribution statement

Peng Wan: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Conceptualization. **Haiyan Xue:** Data curation. **Shukang Zhang:** Validation. **Wentao Kong:** Data curation. **Wei Shao:** Supervision, Conceptualization. **Baojie Wen:** Validation. **Daoqiang Zhang:** Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.media.2025.103557>.

Data availability

Data will be made available on request.

References

- Ambai, M., Yoshida, Y., 2011. CARD: compact and real-time descriptors. In: IEEE International Conference on Computer Vision, ICCV 2011, Barcelona. pp. 97–104.
- Andrew, G., Arora, R., Bilmes, J., Livescu, K., 2013. Deep canonical correlation analysis. In: International Conference on Machine Learning. pp. 1247–1255.
- Bartolotta, T.V., Taibbi, A., Midiri, M., Lagalla, R., 2019. Contrast-enhanced ultrasound of hepatocellular carcinoma: Where do we stand? *Ultrasonography* 38, 200–214.
- Bilen, H., Fernando, B., Gavves, E., Vedaldi, A., Gould, S., 2016. Dynamic image networks for action recognition. In: Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2016. pp. 3034–3042.
- Chadebec, C., Thibaud-Sutre, E., Burgos, N., Allasonnière, S., 2022. Data augmentation in high dimensional low sample size setting using a geometry-based variational autoencoder. *IEEE Trans. Pattern Anal. Mach. Intell.* 45 (3), 2879–2896.
- Chen, C., Wang, Y., Niu, J., Liu, X., Li, Q., Gong, X., 2021. Domain knowledge powered deep learning for breast cancer diagnosis based on contrast-enhanced ultrasound videos. *IEEE Trans. Med. Imaging* 40 (9), 2439–2451.
- Chen, X., Zhou, Y., Wu, D., Zhang, W., Zhou, Y., Li, B., Wang, W., 2022. Imagine by reasoning: A reasoning-based implicit semantic data augmentation for long-tailed classification. In: Thirty-Sixth AAAI Conference on Artificial Intelligence, AAAI 2022, Thirty-Fourth Conference on Innovative Applications of Artificial Intelligence, IAAI 2022, the Twelfth Symposium on Educational Advances in Artificial Intelligence, EAAI 2022 Virtual Event, February 22 - March 1, 2022. AAAI Press, pp. 356–364.
- Cherian, A., Fernando, B., Harandi, M., Gould, S., 2017. Generalized rank pooling for activity recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 3222–3231.
- Cheung, T.-H., Yeung, D.-Y., 2020. Modals: Modality-agnostic automated data augmentation in the latent space. In: International Conference on Learning Representations.
- Dalmaz, O., Saglam, B., Elmas, G., Mirza, M., Çukur, T., 2023. Denoising diffusion adversarial models for unconditional medical image generation. In: 2023 31st Signal Processing and Communications Applications Conference. SIU, pp. 1–5.
- Doersch, C., 2016. Tutorial on variational autoencoders. *CoRR* arXiv:1606.05908.
- Feng, Y., Yang, F., Zhou, X., Guo, Y., Tang, F., Ren, F., Guo, J., Ji, S., 2018. A deep learning approach for targeted contrast-enhanced ultrasound based prostate cancer detection. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 16 (6), 1794–1801.
- Fernando, B., Gavves, E., Oramas, J., Ghodrati, A., Tuytelaars, T., 2016. Rank pooling for action recognition. *IEEE Trans. Pattern Anal. Mach. Intell.* 39 (4), 773–787.
- Frid-Adar, M., Diamant, I., Klang, E., Amitai, M., Goldberger, J., Greenspan, H., 2018. GAN-based synthetic medical image augmentation for increased cnn performance in liver lesion classification. *Neurocomputing* 321, 321–331.
- Gong, R., Danelljan, M., Sun, H., Mangas, J.D., Gool, L.V., 2023. Prompting diffusion representations for cross-domain semantic segmentation. *CoRR* arXiv:2307.02138.
- Guo, L.-H., Wang, D., Qian, Y.-Y., Zheng, X., Zhao, C.-K., Li, X.-L., Bo, X.-W., Yue, W.-W., Zhang, Q., Shi, J., et al., 2018. A two-stage multi-view learning framework based computer-aided diagnosis of liver tumors with contrast enhanced ultrasound images. *Clin. Hemorheol. Microcirc.* 69 (3), 343–354.
- Han, X., Gong, B., Guo, L., Wang, J., Ying, S., Li, S., Shi, J., 2023. B-mode ultrasound based CAD for liver cancers via multi-view privileged information learning. *Neural Netw.* 164, 369–381.
- Han, Z., Yang, F., Huang, J., Zhang, C., Yao, J., 2022a. Multimodal dynamics: Dynamical fusion for trustworthy multimodal classification. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 20707–20717.
- Han, Z., Zhang, C., Fu, H., Zhou, J.T., 2022b. Trusted multi-view classification with dynamic evidential fusion. *IEEE Trans. Pattern Anal. Mach. Intell.* 45 (2), 2551–2566.
- Hong, Y., Zhang, J., Sun, Z., Yan, K., 2022. Safa: Sample-adaptive feature augmentation for long-tailed image classification. In: European Conference on Computer Vision. pp. 587–603.
- Huang, X., Liu, M., Belongie, S.J., Kautz, J., 2018. Multimodal unsupervised image-to-image translation. In: Computer Vision - ECCV 2018 - 15th European Conference, Munich, Germany. In: Lecture Notes in Computer Science, vol. 11207, Springer, pp. 179–196.
- Karras, T., Aittala, M., Laine, S., Härkönen, E., Hellsten, J., Lehtinen, J., Aila, T., 2021a. Alias-free generative adversarial networks. In: Advances in Neural Information Processing Systems. Vol. 34, pp. 852–863.
- Karras, T., Laine, S., Aila, T., 2021b. A style-based generator architecture for generative adversarial networks. *IEEE Trans. Pattern Anal. Mach. Intell.* 43 (12), 4217–4228.
- Li, P., Li, D., Li, W., Gong, S., Fu, Y., Hospedales, T.M., 2021. A simple feature augmentation for domain generalization. In: Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision, ICCV 2021, Montreal, QC, Canada, October 10–17, 2021. IEEE, pp. 8866–8875.

- Li, W., Lin, M.-X., Lin, X.-X., Hu, H.-T., Wang, Y.-C., Ruan, S.-M., Huang, Z.-R., Lu, R.-F., Li, L., Kuang, M., et al., 2024. Systematic comparison of deep-learning based fusion strategies for multi-modal ultrasound in diagnosis of liver cancer. *Neurocomputing* 603, 128257.
- Liang, X., Lin, L., Cao, Q., Huang, R., Wang, Y., 2016. Recognizing focal liver lesions in CEUS with dynamically trained latent structured models. *IEEE Trans. Med. Imaging* 35 (3), 713–727.
- Liu, F., Liu, D., Wang, K., Xie, X., Su, L., Kuang, M., Huang, G., Peng, B., Wang, Y., Lin, M., et al., 2020. Deep learning radiomics based on contrast-enhanced ultrasound might optimize curative treatments for very-early or early-stage hepatocellular carcinoma patients. *Liver Cancer* 9 (4), 397–413.
- Luo, Y., Huang, Q., Liu, L., 2023. Classification of tumor in one single ultrasound image via a novel multi-view learning strategy. *Pattern Recognit.* 143, 109776.
- Minami, Y., Kudo, M., 2011. Review of dynamic contrast-enhanced ultrasound guidance in ablation therapy for hepatocellular carcinoma. *World J. Gastroenterol.* 17, 4952–4959.
- Qin, X., Zhu, J., Tu, Z., Ma, Q., Tang, J., Zhang, C., 2023. Contrast-enhanced ultrasound with deep learning with attention mechanisms for predicting microvascular invasion in single hepatocellular carcinoma. *Academic Radiol.* 30, S73–S80.
- Radzina, M., Ratniece, M., Putrins, D.S., Saule, L., Cantisani, V., 2021. Performance of contrast-enhanced ultrasound in thyroid nodules: Review of current state and future perspectives. *Cancers* 13, 5469.
- Shankar, S., Piratla, V., Chakrabarti, S., Chaudhuri, S., Jyothi, P., Sarawagi, S., 2018. Generalizing across domains via cross-gradient training. In: 6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada. OpenReview.net.
- Su, H., Maji, S., Kalogerakis, E., Learned-Miller, E.G., 2015. Multi-view convolutional neural networks for 3D shape recognition. In: 2015 IEEE International Conference on Computer Vision, ICCV 2015, Santiago, Chile, December 7–13, 2015. pp. 945–953.
- Su, Z., Yao, K., Yang, X., Huang, K., Wang, Q., Sun, J., 2023. Rethinking data augmentation for single-source domain generalization in medical image segmentation. In: Thirty-Seventh AAAI Conference on Artificial Intelligence, AAAI 2023, Washington, DC, USA. AAAI Press, pp. 2366–2374.
- Upchurch, P., Gardner, J.R., Pleiss, G., Pless, R., Snavely, N., Bala, K., Weinberger, K.Q., 2017. Deep feature interpolation for image content changes. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017. IEEE Computer Society, pp. 6090–6099.
- Verma, V., Lamb, A., Beckham, C., Najafi, A., Mitliagkas, I., Lopez-Paz, D., Bengio, Y., 2019. Manifold mixup: Better representations by interpolating hidden states. In: Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9–15 June 2019, Long Beach, California, USA. In: Proceedings of Machine Learning Research, vol. 97, PMLR, pp. 6438–6447.
- Volpi, R., Namkoong, H., Sener, O., Duchi, J.C., Murino, V., Savarese, S., 2018. Generalizing to unseen domains via adversarial data augmentation. In: Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, Montréal, Canada. pp. 5339–5349.
- Wang, W., Arora, R., Livescu, K., Bilmes, J., 2015. On deep multi-view representation learning. In: International Conference on Machine Learning. pp. 1083–1092.
- Wang, X., Girshick, R., Gupta, A., He, K., 2018. Non-local neural networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 7794–7803.
- Wang, Y., Pan, X., Song, S., Zhang, H., Huang, G., Wu, C., 2019. Implicit semantic data augmentation for deep networks. In: Proceedings of the 32 Th Advances in Neural Information Processing Systems: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, Vancouver, BC, Canada. pp. 12614–12623.
- Xu, C., Si, J., Guan, Z., Zhao, W., Wu, Y., Gao, X., 2024. Reliable conflictive multi-view learning. In: Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2024, February 20–27, 2024, Vancouver, Canada. pp. 16129–16137.
- Zhang, H., Guo, L., Wang, J., Ying, S., Shi, J., 2023a. Multi-view feature transformation based svm+ for computer-aided diagnosis of liver cancers with ultrasound images. *IEEE J. Biomed. Heal. Inform.* 27 (3), 1512–1523.
- Zhang, L., Wang, X., Yang, D., Sanford, T., Harmon, S.A., Turkbey, B., Roth, H., Myronenko, A., Xu, D., Xu, Z., 2019. When unseen domain generalization is unnecessary? Rethinking data augmentation. *CoRR arXiv:1906.03347*.
- Zhang, Q., Wu, H., Zhang, C., Hu, Q., Fu, H., Zhou, J.T., Peng, X., 2023b. Provable dynamic fusion for low-quality multimodal data. *arXiv preprint arXiv:2306.02050*.
- Zhao, Q.-x., He, X.-l., Wang, K., Cheng, Z.-g., Han, Z.-y., Liu, F.-y., Yu, X.-l., Hui, Z., Yu, J., Chao, A., et al., 2023. Deep learning model based on contrast-enhanced ultrasound for predicting early recurrence after thermal ablation of colorectal cancer liver metastasis. *Eur. Radiol.* 33 (3), 1895–1905.
- Zhao, Q., Liu, Z., Adeli, E., Pohl, K.M., 2021. Longitudinal self-supervised learning. *Med. Image Anal.* 71, 102051.